

Units in CSS

Understand absolute and relative CSS units for consistent, responsive, and scalable web layouts.

What Are CSS Units?

CSS units define the size, spacing, and positioning of elements on a web page. Understanding units is essential for building layouts that are consistent, responsive, and easy to manage.

Two Categories of Units

1. Absolute Units

These do not change based on screen size or parent element. Use them for fixed-size elements, but use them cautiously in responsive designs.

Unit	Description
px	Pixels — the most common absolute unit.
pt	Points — 1/72 of an inch.
cm	Centimeters.
mm	Millimeters.
in	Inches.

Example

```
h1 {  
  font-size: 24px;  
}
```

2. Relative Units

Relative units are responsive and scale based on parent elements, the root font size, or the viewport size.

Unit	Description
%	Relative to the parent element.
em	Relative to the parent's font size.
rem	Relative to the root font size (usually the <code><html></code> element).
vw	1% of the viewport width.
vh	1% of the viewport height.
vmin	1% of the smaller viewport dimension.
vmax	1% of the larger viewport dimension.

Commonly Used Units

Practical examples of units used frequently in everyday CSS.

px (Pixels)

A fixed unit that does not scale with the screen size. It is useful when you need a specific, controlled size.

% (Percentage)

Useful for making widths or heights relative to the parent element.

```
p {
  margin: 10px;
}

div {
  width: 80%;
}
```

em vs rem

Unit	Relative to
em	The font size of the parent element.
rem	The font size of the root (html) element.

Use **rem** for consistency in modern responsive design.

vw and vh

These units are powerful for creating fullscreen layouts and sizing elements relative to the viewport.

```
div {
  font-size: 2em; /* 2 times the parent's font size */
}

html {
  font-size: 16px;
}

h1 {
  font-size: 2rem; /* 32px */
}

.container {
  width: 100vw; /* Full width of the viewport */
  height: 100vh; /* Full height of the viewport */
}
```

Responsive CSS Units

Choose units according to whether a value should be fixed, relative, or viewport-based.

Quick Comparison

Unit	Relative to	Typical purpose
px	Fixed size	Precise, fixed spacing and sizing.
%	Parent element	Flexible widths and heights.
em	Parent font size	Sizing relative to local typography.
rem	Root font size	Consistent scalable typography.
vw	Viewport width	Viewport-based widths and layouts.
vh	Viewport height	Viewport-based heights and fullscreen sections.
vmin	Smaller viewport dimension	Sizing based on the smaller viewport dimension.
vmax	Larger viewport dimension	Sizing based on the larger viewport dimension.

calc() Function

The **calc()** function lets you combine different CSS units in a single calculation.

```
section {  
  width: calc(100% - 200px);  
}
```

■ Tip

Think about what a value should respond to before choosing its unit: the element's parent, the root font size, or the viewport.

Best Practices & Summary

A compact reference for choosing CSS units in real layouts.

Best Practices

- Use **rem** for typography for consistency and scalability.
- Use **%**, **vw**, and **vh** for responsive layouts.
- Avoid overusing **px** in responsive designs.

Unit Selection Guide

When you need...	Consider...
A fixed value	px or another absolute unit.
A value relative to a parent	% or em .
Consistent scalable typography	rem .
Viewport-based sizing	vw or vh .
Responsive viewport extremes	vmin or vmax .
A calculated combination	calc() .

Quick Reference

Category	Units
Absolute	px, pt, cm, mm, in
Relative	%, em, rem, vw, vh, vmin, vmax

Quick Takeaway

CSS units help control the size and spacing of elements. Choosing the right unit is key to making layouts flexible, scalable, and consistent across devices.